SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool:	Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: **Total Marks: 10**

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

```
A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
```

```
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>
```

Snippet 2: Form Submission Methods

```
A. GET
B. POST
C. PUT
D. DELETE
```

Snippet 3: Navigation Elements

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

Snippet 4: XML Syntax

```
A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note
```

Snippet 5: Web Communication Protocols

```
A. FTP
B. HTTP
C. SMTP
D. SNMP
```

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

*I (*B.Manikanteswara Reddy*/192372225) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

B.Manikanteswara Reddy
Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Scenario: Web Development Standards Implementation

Snippet 1: Video Embedding Options

A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>

1. Select the correct HTML tag for video embedding and justify your answer

Answer: B. <video src="video.mp4" controls></video>
Justification:

- <video> is the official HTML5 element for embedding videos.
- The controls attribute displays playback controls.
- It supports formats like MP4, WebM, and Ogg.

Why the other options are incorrect:

- **A. <media>** – Not a valid HTML element.
- **C. <movie>** – Not an HTML tag.
- **D. <vid>** – Not a valid HTML tag.

Outcome: The promotional video will display correctly with user controls.

Snippet 2: Form Submission Methods

A. GET
B. POST
C. PUT
D. DELETE

2. Identify the appropriate HTTP method for form submission and explain why

Answer: B. POST
Justification:
Sends data in the request body.
Suitable for registration forms containing sensitive information.
Supports larger amounts of data than GET.
Why the other options are incorrect:
GET – Sends data in the URL, making it visible and less secure.
PUT – Primarily used to update existing resources.
DELETE – Used to remove resources from the server.
Outcome: User registration data is transmitted more securely and appropriately.

Snippet 3: Navigation Elements

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

3. Choose the proper semantic element for navigation and justify its usage
Answer: B. <nav>
Justification:
Represents a section containing navigation links.
Improves accessibility for screen readers.

Helps search engines understand page structure.
Why the other options are incorrect:
<div> – Generic container with no semantic meaning.
<section> – Used for grouping related content, not specifically navigation.
<header> – Represents introductory content but is not dedicated to navigation.
Outcome: The website follows semantic HTML standards and is easier to navigate

Snippet 4: XML Syntax

A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note

4. **Identify the correct XML syntax and explain the error in other options.**

Answer: B. <note><to>User</to><from>Admin</from></note>
Justification:
Every opening tag has a matching closing tag.
Tags are properly nested.
The document is well-formed XML.
Errors in other options:
A. Missing closing </note> tag.
C. Incorrect closing tag (</from> instead of </to>).
D. Missing closing angle bracket (>) for the final tag.
Outcome: The XML configuration file can be parsed successfully without errors.

Snippet 5: Web Communication Protocols

A. FTP
B. HTTP
C. SMTP
D. SNMP

5. **Choose the correct protocol for web communication and justify your choice.**

Answer: B. HTTP
Justification:
HTTP is the standard protocol used for communication between web browsers and web servers.
Modern websites typically use **HTTPS**, which adds encryption through SSL/TLS.
Why the other options are incorrect:
FTP – Used for file transfer.
SMTP – Used for sending email.
SNMP – Used for network monitoring and management.
Outcome: The web application communicates correctly between the client and server using the standard web protocol.